Tabish Izhar

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Profile

Software engineer with about 2 years of industry experience in game development, backend development, and IT support, now pursuing an M.Sc. in International Software Systems Science at the University of Bamberg. Focused on moving into AI/ML engineering, with hands-on coursework in machine learning, deep learning, NLP, data streams, and cloud-native systems (Docker, Kubernetes, microservices). Comfortable working in Scrum teams and taking features from idea to release.

Education

M.Sc. International Software Systems Science (ISoSySc) 2024 – Present
University of Bamberg, Germany

Relevant modules and work:

- **Service-Oriented Architecture (SOA):** Studied microservices, Docker and Kubernetes; built a web service and REST APIs (Java, Jersey) in a microservices project.
- **ADAML (Data Analytics & Machine Learning):** Data analysis and ML methods including linear regression, logistic regression and k-nearest neighbours (KNN).
- **Deep Learning for NLP:** Tokenization, text representation and neural approaches to natural language processing.
- **Project – Dialogue Systems (DS Conv AI):** Worked on the Natural Language Understanding (NLU) component of a conversational AI chatbot.
- **Data Stream Processing (DSC):** Concepts and techniques for processing continuous data streams.
- **Evidence-Based Software Engineering:** Wrote a research paper analysing security vulnerabilities in code generated by different AI models.
- **DT-CPP-M (C++ Programming):** Solved a large set of C++ assignments, ensuring every solution passed all automated test cases.
- **DBCPU-M (Databases):** In-depth study and practice of SQL queries; PostgreSQL work on the Enron dataset.

B.Sc. Computer Science 2017 – 2022
NCBAE, Pakistan

- Teaching Assistant for Programming Fundamentals, Digital Logic Design and Discrete Structures.

Professional Experience

Software Engineer 2023 – 2024
Gaminations, Lahore, Pakistan

- Worked on full-cycle development using C++, Python, JavaScript, PHP and MySQL for high-performance, feature-rich game applications.
- Developed automation scripts in Shell and Python to streamline Git operations and validate JSON-based game levels against acceptance criteria.
- Contributed to gameplay design with FairyGUI, Cocos2d-x and Cocos Creator, implementing features such as PowerUps, PopUps and tutorials to increase user engagement.
- Maintained and improved backend logic in PHP to meet changing product requirements.
- Managed Hotfix and Release branches for clean deployment workflows.
- Designed and implemented UI/UX elements using FGUI and JSON configurations for new features.
- Monitored app performance through Firebase Console and Google Play Console; tracked and reported bugs in Atlassian Jira.
- Worked in a Scrum team using Jira, Confluence and SourceTree for tasks, sprints and collaboration.

IT & Digital Workflow Specialist 2022 – 2023
ClearPath Orthodontics, Lahore, Pakistan

- Worked with 3Shape OrthoAnalyzer and Dent One for orthodontic case analysis and digital workflows.
- Provided IT support for hardware, software and network troubleshooting within the department.
- Collaborated with clinical teams to optimise digital dental processes.
- Managed the department's IT support for one month, handling desktop and network troubleshooting.

Projects

Linear Regression from Scratch github.com/tabishizhar9

- Built linear regression with my own gradient descent implementation on the California Housing dataset.
- Served the model through a FastAPI backend, containerised with Docker and deployed live on Render.

Certifications & Ongoing Learning

- Machine Learning Specialization – DeepLearning.AI / Stanford (Coursera), in progress
- Deep Learning Specialization – DeepLearning.AI (Coursera), in progress

Technical Skills

Languages:	Python, C++, JavaScript, PHP, Java, SQL, Shell
ML / Data:	Linear & logistic regression, KNN, gradient descent, NumPy, NLP basics (tokenization)
Backend / Web:	FastAPI, REST APIs, Jersey, MySQL, PostgreSQL
DevOps / Cloud:	Docker, Kubernetes, microservices, Git, branching & release management
Game Dev:	Cocos2d-x, Cocos Creator, FairyGUI (FGUI)
Monitoring:	Firebase Console, Google Play Console
Tools:	Jira, Confluence, SourceTree
Domain tools:	3Shape OrthoAnalyzer, Dent One

Languages

English – Fluent | Urdu – Native | German – [Level, e.g. A2/B1]